

# Dr Waseem Raza, PhD

## Machine Learning Engineer | AI/ML, GenAI & Wireless Systems

Applied AI | Generative & Agentic AI | Data Science | Wireless Networks

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### Professional Summary

Machine Learning Engineer with a PhD in Electrical and Computer Engineering and experience spanning applied AI, Generative AI, data science, wireless-network intelligence, and production-oriented engineering. Work includes diagnostic machine learning, domain adaptation, AI-ready data pipelines, model evaluation, cloud analytics, intelligent automation, and research in resilient wireless systems. Combines industry delivery at AT&T and Impac Exploration Services with research experience at AT&T Labs, Futurewei, and the University of Oklahoma.

### Technical Expertise, Leadership, and Professional Strengths

**Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, classification, regression, anomaly detection, transfer learning, domain adaptation, time-series modeling, cross-validation, and model evaluation

**Generative & Agentic AI:** RAG, embeddings, vector retrieval, prompt engineering, structured extraction, tool-using agents, LangChain, LangGraph, agent orchestration, and grounded response generation

**Data Engineering & Analytics:** Python, SQL, Pandas, NumPy, ETL/ELT, feature engineering, data validation, Azure Databricks, Apache Spark, PySpark, and cloud analytics workflows

**MLOps & Software Engineering:** Docker, Kubernetes, OpenShift, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, model deployment, and reproducible workflows

**Wireless Systems:** LTE/4G, 5G-NR, NR-NTN, mobility, RAN optimization, propagation modeling, network measurements, and service assurance

**Wireless AI & Reliability:** Network intelligence, outage management, transfer learning, adversarial learning, generative augmentation, self-healing, and resilient automation

#### Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

### Professional Experience

#### Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

#### Data Scientist and AI/ML Engineer

Impac Exploration Services

Sep. 2024-Dec. 2024

Weatherford, OK

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Advanced-Degrees Intern</b><br><i>AT&amp;T Labs</i> <ul style="list-style-type: none"> <li>Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.</li> <li>Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.</li> <li>Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Jun. 2024-Aug. 2024</b><br><i>Austin, TX</i>     |
| <b>Advanced Wireless Technology Development Intern</b><br><i>Futurewei Technologies</i> <ul style="list-style-type: none"> <li>Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.</li> <li>Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.</li> <li>Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Sep. 2022-Jan. 2023</b><br><i>Schaumburg, IL</i> |
| <b>Advanced-Degrees Intern</b><br><i>AT&amp;T Labs</i> <ul style="list-style-type: none"> <li>Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.</li> <li>Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.</li> <li>Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Jun. 2022-Aug. 2022</b><br><i>Bedminster, NJ</i> |
| <b>Graduate Research Assistant</b><br><i>AI4Networks, University of Oklahoma</i> <ul style="list-style-type: none"> <li>Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.</li> <li>Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.</li> <li>Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.</li> <li>Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.</li> <li>Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.</li> </ul> | <b>Sep. 2019-Sep. 2024</b><br><i>Tulsa, OK</i>      |

Education

|                                                                                                         |                 |
|---------------------------------------------------------------------------------------------------------|-----------------|
| <b>PhD, Electrical and Computer Engineering</b><br>University of Oklahoma, Norman, Oklahoma, USA        | 09/2019-07/2025 |
| <b>MSc, Telecommunication Engineering</b><br>University of Engineering and Technology, Taxila, Pakistan | 08/2014-08/2016 |
| <b>BSc, Telecommunication Engineering</b><br>University of Engineering and Technology, Taxila, Pakistan | 11/2010-06/2014 |

Projects

|                                                                                                                                                                                                                                                                                                                                                                      |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Agentic AI and RAG Applications</b><br><i>Portfolio Project</i> <ul style="list-style-type: none"> <li>Practical experiments with retrieval, tool-using agents, orchestration, memory, evaluation, and multi-agent workflows.</li> </ul> <i>Tools &amp; methods:</i> LangGraph, CrewAI, AutoGen, RAG, Vector Databases                                            | 2026 / In progress |
| <b>AI-Ready Lakehouse Pipeline</b><br><i>Portfolio Project</i> <ul style="list-style-type: none"> <li>An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads.</li> </ul> <i>Tools &amp; methods:</i> Azure, Databricks, Spark, Data Engineering                                                                          | 2024               |
| <b>SyntheticNET: 3GPP-Compliant Network Simulation</b><br><i>Research Project</i> <ul style="list-style-type: none"> <li>Python-based simulation environment for realistic mobility, propagation, and data-driven experimentation in cellular networks.</li> </ul> <i>Tools &amp; methods:</i> Python, 3GPP, Simulation, Wireless Networks                           | 2020-2024          |
| <b>Wearable Intelligence for Health Monitoring</b><br><i>Research Project</i> <ul style="list-style-type: none"> <li>Deep-learning methods for continuous recognition of medication and dietary activities from wrist-worn inertial sensors.</li> </ul> <i>Tools &amp; methods:</i> Wearable AI, Time Series, Deep Learning, Smartwatch Sensing                      | 2024-2026          |
| <b>System Design Case-Study Portfolio</b><br><i>Guided Project</i> <ul style="list-style-type: none"> <li>Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs.</li> </ul> <i>Tools &amp; methods:</i> System Design, APIs, Distributed Systems, Scalability | 2026 / In progress |

Certifications and Continuing Education

1. AWS Generative AI and AI Agents with Amazon Bedrock — [AWS/Coursera](#) | Completed (3/3) courses | 08/2026 | [Certificate](#)

**Focus:** Three-course professional certificate focused on developing, customizing, optimizing, and automating generative AI applications and intelligent agents with Amazon Bedrock.

**Tools & Skills:** Amazon Bedrock, Generative AI, AI Agents, Foundation Models, Knowledge Bases, Retrieval-Augmented Generation, LangChain, Amazon Q Developer, Model Customization, Prompt Engineering, Natural Language Processing, Text Generation, Summarization, Generative AI Application Development
2. IBM RAG and Agentic AI Professional Certificate — [IBM / Coursera](#) | Completed (8/8) courses | February 2026 | [Certificate](#)

Original eight-course credential.

**Focus:** Completed eight-course professional certificate covering retrieval-augmented generation, vector databases, multimodal applications, and agentic AI workflows.

**Tools & Skills:** Retrieval-Augmented Generation, Agentic AI, LangChain, LangGraph, Vector Databases, ChromaDB, FAISS, LlamaIndex, Multimodal AI, Tool Calling, Prompt Engineering, CrewAI, AutoGen / AG2, BeeAI, Gradio, Python, Flask
3. Machine Learning — Stanford University / Coursera | Completed

**Focus:** Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

**Tools & Skills:** Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization
4. Google Cloud Data Analytics Professional Certificate — Google Cloud / Coursera | Completed

**Focus:** Cloud data transformation, analysis, visualization, and decision support using Google Cloud tools.

**Tools & Skills:** SQL, BigQuery, Data Analytics, Visualization
5. AI for Telecommunications Specialization — [AI CERTs / Coursera](#) | Completed (3/3) courses | June 2026 | [Certificate](#)

**Focus:** Three-course specialization connecting AI foundations with telecommunications network optimization, security, and customer experience.

**Tools & Skills:** AI for Telecommunications, Network Optimization, Cybersecurity, Customer Experience, Internet of Things, Python, TensorFlow
6. 4G Network Fundamentals — [Institut Mines-Télécom / Coursera](#) | Completed (1/1) courses | August 2026 | [Certificate](#)

**Focus:** Foundational study of fourth-generation cellular networks.

**Tools & Skills:** 4G Networks, Cellular Networks, Telecommunications

Publications

|                                                                                                     |      |
|-----------------------------------------------------------------------------------------------------|------|
| <b>Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches</b>     | 2026 |
| <i>HealthINF   Accepted</i>                                                                         |      |
| Wearable AI for continuous, real-world medication-intake monitoring.                                |      |
| <b>On Multi-Parameter Optimization and Proactive Reliability in Cellular Networks</b>               | 2025 |
| <i>Journal article</i>                                                                              |      |
| Data-driven optimization and proactive reliability for future cellular-network operation.           |      |
| <b>An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin</b> | 2024 |
| <i>IEEE PIMRC</i>                                                                                   |      |
| Robust, data-driven propagation modeling for wireless-network digital twins.                        |      |